

2025

COMPUTER SCIENCE — HONOURS

Paper : SEC-3

(Mobile App Development)

Full Marks : 50

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*Answer **question no. 1** and **any four** questions from the rest.1. Answer **any five** questions :

2×5

- (a) What are the alternative methods for running Flutter?
- (b) How does an IDE differ from an SDK?
- (c) What is material design? Explain with appropriate illustrations.
- (d) What are assets related to flutter?
- (e) What is a List View widget?
- (f) What is Navigator? Explain in brief.
- (g) What do you mean by widget Tree?
- (h) What is const constructor? What is the purpose of using it in Flutter?

2. (a) What is pubspec.yaml file? What is its purpose?

(b) What are Runes? Explain the use of it with suitable examples.

4+6

3. (a) Differentiate between a full widget tree and a shallow widget tree in Flutter. Explain with an example how a full widget tree is built.

(b) Explain how complex UIs are created using nested widgets in Flutter. Draw a simple diagram showing a widget tree for a small app.

(3+2)+(3+2)

4. (a) What are Maps in Dart? Explain how to declare, initialize, and use Maps with suitable examples.

(b) What do you mean by Native and Cross platform Development?

(c) Explain how functions and classes are used in Dart programs.

(2+2)+3+3

5. What is JSON? Why is JSON commonly used in Web Development? Explain the overall structure of JSON.

3+3+4

6. (a) How is local storage created in Flutter? Explain with proper illustrations.

(b) What is Firebase storage? Explain with suitable example.

(3+2)+(3+2)

Please Turn Over

(5267)

7. (a) What is the purpose of container? Explain with appropriate illustrations.

(b) What is state management and how is it handled in Flutter?

2+3+5

8. Write short notes on (*any two*) :

5×2

(a) Stateless widget life cycle

(b) Hero Animation

(c) Android Virtual Device (AVD) and its use

(d) Asynchronous programming with respect to Dart.
